LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 2 Test (GSEB pattern): Acids, Bases and Salts

Chapter / Topic: Acids, Bases and Salts

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. A solution turns red litmus blue. Its pH is most likely to be: **[1]**

(a) 1 (b) 4
(c) 5 (d) 10

Q2. Which of the following gives carbon dioxide with dilute hydrochloric acid? **[1]**

(a) Sodium hydrogencarbonate (b) Zinc
(c) Sodium hydroxide (d) Sodium chloride

Q3. Which gas is given off when dilute hydrochloric acid is added to sodium carbonate? **[1]**

(a) Hydrogen (b) Carbon dioxide
(c) Chlorine (d) Oxygen

Q4. Which of these has the highest pH? **[1]**

(a) Lemon juice (b) Pure water
(c) Milk of magnesia (d) Vinegar

Q5. The chemical formula of bleaching powder is: **[1]**

(a) $CaCO_3$ (b) $Ca(OH)_2$
(c) $CaOCl_2$ (d) $CaSO_4$

Q6. Which indicator turns pink in a basic solution? **[1]**

(a) Methyl orange (b) Blue litmus
(c) Turmeric (d) Phenolphthalein

Q7. Which of the following is used as an antacid? **[1]**

(a) Sodium chloride (b) Milk of magnesia
(c) Vinegar (d) Lemon juice

Q8. The chemical formula of Plaster of Paris is: **[1]**

(a) $CaSO_4 \cdot \frac{1}{2}H_2O$ (b) $CaSO_4 \cdot 2H_2O$
(c) $CaSO_4$ (d) $CaCO_3$

Q9. Plaster of Paris is obtained by heating: **[1]**

(a) limestone (b) baking soda
(c) gypsum (d) washing soda

SECTION B

6 marks

Q10. Why does dry HCl gas not change the colour of dry blue litmus paper? **[2]**

Q11. What is a universal indicator? Why is it more useful than litmus? **[2]**

Q12. Write the chemical name and formula of baking soda. Write the equation for the reaction that takes place when it is heated. **[2]**

SECTION C

6 marks

Q13. A student tests three solutions P, Q and R with universal indicator and finds their pH to be 2, 7 and 12 respectively. **[3]**

(i) Which solution is neutral? **[1]**

(ii) Which solution has the highest concentration of H^+ ions? **[1]**

(iii) Which solution will turn phenolphthalein pink? **[1]**

Q14. A solution has a pH of 3. **[3]**

(i) Is the solution acidic, basic or neutral? **[1]**

(ii) What happens to its pH when it is diluted with water? **[1]**

(iii) What happens to the pH if a base is added slowly until the solution is neutralised? **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. Tooth enamel, made of calcium hydroxyapatite (a form of calcium phosphate), is the hardest substance in the body. It does not dissolve in water, but it is corroded when the pH in the mouth falls below 5.5. Bacteria in the mouth produce acids by breaking down sugar and food particles that remain after eating. Toothpastes, which are generally basic, are used to clean the teeth. **[4]**

(i) Below what pH does tooth decay start? **[1]**

(ii) Why are toothpastes generally basic? **[1]**

(iii) What produces the acid in the mouth? Suggest two ways to prevent tooth decay. **[2]**

~ All the best! ~